

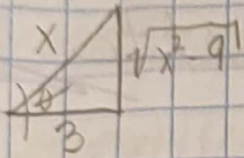
Tarea Sección 7.3

(3secθ tanθ) dθ

$$1) \int \frac{1}{x^2 \sqrt{x^2 - 9}} dx = \int \frac{1}{x^2 \sqrt{x^2 - 3^2}} = \int \frac{1}{(3\sec\theta)^2 \sqrt{(3\sec\theta)^2 - 3^2}} \cdot (3\sec\theta \tan\theta) d\theta$$

$$x = 3\sec\theta \quad dx = 3\sec\theta \tan\theta d\theta$$

$$\frac{x}{3} = \sec\theta \quad = \frac{3}{3} \int \frac{\sec\theta \tan\theta}{9\sec^2\theta \sqrt{9\sec^2\theta - 9}} d\theta$$



$$= \frac{1}{9} \int \frac{\sec\theta \tan\theta}{\sec^2\theta \sqrt{\sec^2\theta - 1}} d\theta$$

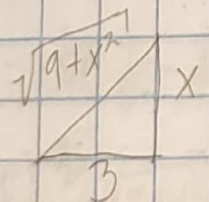
$$= \frac{1}{9} \int \frac{\sec\theta \tan\theta}{\sec^2\theta \sqrt{\tan^2\theta}} = \frac{1}{9} \int \frac{\sec\theta \tan\theta}{\sec^2\theta \tan\theta} = \frac{1}{9} \int \frac{1}{\sec\theta} d\theta = \frac{1}{9} \int \cos\theta d\theta$$

$$\frac{1}{9} \sin\theta + C = \frac{1}{9} \left(\frac{\sqrt{x^2 - 9}}{x} \right) + C = \frac{\sqrt{x^2 - 9}}{9x} + C$$

$$3) \int \frac{x^3}{\sqrt{x^2 + 9}} dx = \int \frac{x^3}{\sqrt{3^2 + x^2}} dx = \int \frac{(3\tan\theta)^3}{\sqrt{(3\tan\theta)^2 + 3^2}} \cdot 3\sec^2\theta d\theta$$

$$x = 3\tan\theta \quad dx = 3\sec^2\theta d\theta$$

$$\frac{x}{3} = \tan\theta$$



$$= \int \frac{27\tan^3\theta}{\sqrt{9\tan^2\theta + 9}} \cdot 3\sec^2\theta d\theta$$

$$= \frac{81}{3} \int \frac{\tan^3\theta \sec^2\theta}{\sqrt{\tan^2\theta + 1}} d\theta = 27 \int \frac{\tan^3\theta \sec^2\theta}{\sec\theta} d\theta$$

$$= 27 \int \tan^2\theta \sec\theta d\theta = 27 \int \tan\theta \tan^2\theta \sec\theta d\theta$$

$$= 27 \int \tan\theta (\sec^2\theta - 1) \sec\theta d\theta = 27 \int \tan(m^2 - 1) m dm$$

$$m = \sec\theta$$

$$dm = \sec\theta \tan\theta d\theta$$

$$= 27 \int m^2 - 1 dm$$

$$= 27 \left(\frac{m^3}{3} - m \right) = 27 \left(\frac{\sec^3\theta}{3} - \sec\theta \right) + C$$

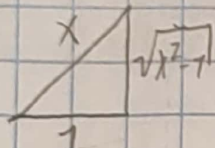
$$= 27 \left[\frac{1}{3} \left(\frac{\sqrt{9 + x^2}}{3} \right)^3 - \left(\frac{\sqrt{9 + x^2}}{3} \right) \right]$$

$$5) \int \frac{\sqrt{x^2-1}}{x^4} dx = \frac{\sqrt{x^2-1}}{x^4} \cdot \frac{\sqrt{x^2-1}}{\sqrt{x^2-1}} = \int \frac{x^2-1}{x^4 \sqrt{x^2-1}} dx = \int \frac{\sec^2 \theta - 1}{(\sec \theta)^4 \sqrt{(\sec^2 \theta - 1)}} \cdot \sec \theta \tan \theta d\theta$$

$$x = \sec \theta \quad dx = \sec \theta \tan \theta d\theta$$

$$\frac{x}{1} = \sec \theta$$

$$= \int \frac{\tan^2 \theta \sec \theta \tan \theta}{\sec^4 \theta \sqrt{\tan^2 \theta}} d\theta = \int \frac{\tan^2 \theta}{\sec^3 \theta} d\theta = \int \frac{\sin^2 \theta}{\cos^3 \theta} d\theta$$



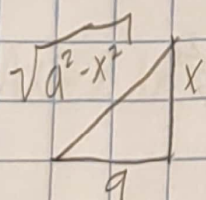
$$m = \sin \theta$$

$$dm = \cos \theta d\theta$$

$$7) \int_0^a \frac{dx}{(a^2+x^2)^{3/2}} = \int_0^{\pi/4} \frac{\sec^2 \theta d\theta}{(a^2 + a^2 \tan^2 \theta)^{3/2}} = \int_0^{\pi/4} \frac{\sec^2 \theta d\theta}{(a^2 \sec^2 \theta)^{3/2}} = \frac{1}{a^2} \int_0^{\pi/4} \sec^2 \theta d\theta$$

$$x = a \tan \theta$$

$$\frac{x}{a} = \tan \theta$$



$$dx = a \sec^2 \theta d\theta$$

$$= \int \frac{1}{\sec \theta} d\theta = \int \cos \theta d\theta = \frac{1}{a} (\sin \theta) \Big|_0^{\pi/4}$$

$$\frac{x}{\sqrt{a^2+x^2}} = \frac{1}{a^2} \left[\frac{a}{\sqrt{a^2-a^2}} \right] - \left[\frac{0}{\sqrt{a^2-0}} \right] =$$

$$dx = a$$

$$\theta = \tan^{-1} \left(\frac{x}{a} \right)$$

$$\theta = \tan^{-1} \left(\frac{a}{a} \right) = \frac{\pi}{4}$$

$$dx = 0$$

$$\theta = \tan^{-1} \left(\frac{0}{a} \right)$$

$$\theta = \tan^{-1}(0) = 0$$

$$\sqrt{a^2 + (a \tan \theta)^3}$$

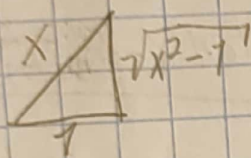
$$\sqrt{a^2 + a \tan \theta}^3$$

$$\sqrt{a(a + \tan \theta)}^3$$

$$9) \int_2^3 \frac{dx}{(x^2-1)^{3/2}} = \int \frac{dx}{\sqrt{(x^2-1)^3}} = \int \frac{\sec\theta \tan\theta}{\sqrt{\sec^2 x - 1}^3} d\theta = \int \frac{\sec\theta \tan\theta}{\tan^3 \theta} d\theta$$

$$x = \sec\theta$$

$$\frac{x}{1} = \sec\theta$$



$$dx = \sec\theta \tan\theta d\theta$$

$$= \int \frac{\sec\theta}{\tan^2 \theta} d\theta = \int \frac{\sec\theta}{\sin^2 \theta} d\theta$$

$$u = \sec\theta$$

$$du = \sec\theta \tan\theta d\theta$$

$$= \int \frac{1}{u^2} \frac{du}{\cos\theta} = \int \frac{1}{u^2} du = -\frac{1}{u} = -\frac{1}{\sec\theta} = -\cos\theta = -\frac{1}{x} \Big|_2^3$$

$$= -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}$$

$$11) \int_0^{1/2} x \sqrt{1-4x^2} dx = \int \frac{x \sqrt{m}}{-8x} dm = -\frac{1}{8} \int \sqrt{m} dm = -\frac{1}{8} \cdot \frac{2}{3} m^{3/2} \Big|_0^{1/2} = -\frac{1}{12} \left(\frac{1}{2} \right)^{3/2} = -\frac{1}{12}$$

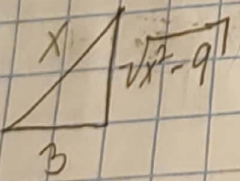
$$m = 1-4x^2$$

$$dm = -8x dx$$

$$13) \int \frac{\sqrt{x^2-9}}{x^3} dx = \int \frac{\sqrt{x^2-9}}{x^3 \sqrt{x^2-9}} dx = \int \frac{1}{x^3} dx = -\frac{1}{2x^2} + C$$

$$x = 3\sec\theta$$

$$\frac{x}{3} = \sec\theta$$



$$dx = \sec\theta \tan\theta d\theta$$

$$= \int \frac{1}{(3\sec\theta)^3} \sec\theta \tan\theta d\theta = \int \frac{\tan\theta}{27 \sec^2 \theta} d\theta = \frac{1}{27} \int \frac{\tan\theta}{\sec^2 \theta} d\theta$$

$$= \frac{1}{27} \int \frac{\sin\theta}{\cos^2 \theta} d\theta = \frac{1}{27} \int \frac{\sin\theta \cos\theta}{\cos^2 \theta} d\theta = \frac{1}{27} \int \frac{\sin\theta}{\cos\theta} d\theta = -\frac{1}{27} \ln|\cos\theta| + C$$

$$= -\frac{1}{27} \ln\left|\frac{1}{\sec\theta}\right| = \frac{1}{27} \ln|\sec\theta| = \frac{1}{27} \ln\left|\frac{x}{3}\right| + C$$

$$= \frac{1}{27} \ln\left|\frac{\sqrt{x^2-9}}{3}\right| + C = \frac{1}{27} \ln|\sqrt{x^2-9}| + C$$

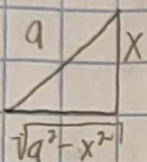
$$m = \sin\theta$$

$$dm = \cos\theta d\theta$$

$$15) \int_0^a x^2 \sqrt{a^2 - x^2} dx = \int a^2 \sin^2 \theta \sqrt{a^2 - a^2 \sin^2 \theta} a \cos \theta d\theta$$

$$x = a \sin \theta \quad dx = a \cos \theta d\theta$$

$$\frac{x}{a} = \sin \theta$$



$$= a^3 \int \sin^2 \theta \sqrt{1 - \sin^2 \theta} \cos \theta d\theta$$

$$= a^3 \int \sin^2 \theta \cos^2 \theta \cos \theta d\theta$$

$$= a^4 \int \sin^2 \theta (1 - \sin^2 \theta) d\theta = a^4 \int \sin^2 \theta - \sin^4 \theta d\theta$$

$$m = 2\theta \\ dm = 2d\theta$$

$$= a^4 \left[\frac{1 - \cos 2\theta}{2} d\theta - \int \sin^2 \theta \sin^2 \theta d\theta \right]$$

$$\sin^{-1}(1) = \frac{\pi}{2}$$

$$\sin^{-1}(0) = 0$$

$$= a^4 \left[\frac{1}{2} \int (1 - \cos 2\theta) d\theta - \int \sin^2 \theta \sin^2 \theta d\theta \right]$$

$$= a^4 \left[\frac{1}{2} \int (1 - \cos m) d\theta - \int (1 - \cos m) d\theta (1 - \cos m) d\theta \right]$$

$$= a^4 \left[\frac{1}{4} (\theta - \sin 2\theta) - \frac{1}{2} (\theta - \sin 2\theta) \frac{1}{2} (\theta - \sin 2\theta) \right]$$

$$= a^4 \left[\frac{1}{4} \theta - \frac{1}{10} \sin 2\theta + \frac{1}{4} \sin^3 \theta \cos \theta \right] \Big|_0^{\pi/2}$$

$$= a^4 \left[\frac{1}{8} \theta - \frac{1}{10} \sin 2\theta + \frac{1}{4} \sin^3 \theta \cos \theta \right]$$

$$= a^4 \left[\frac{1}{8} \sin^{-1} \left(\frac{x}{a} \right) - \frac{1}{10} (2 \sin \theta \cos \theta) + \frac{1}{4} \left(\frac{x^3}{a^3} \right) \left(\frac{\sqrt{a^2 - x^2}}{a} \right) \right] \Big|_0^{\pi/2}$$

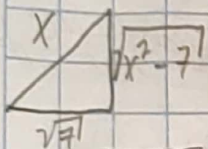
$$= a^4 \left[\frac{1}{8} \sin^{-1} \left(\frac{1}{2} \right) - \frac{1}{8} \left(\frac{x}{a} \right) \left(\frac{\sqrt{a^2 - x^2}}{a} \right) + \frac{1}{4} \left(\frac{\pi^3}{a^3} \right) \left(\frac{\sqrt{a^2 - \frac{\pi^2}{4}}}{a} \right) \right] = \frac{7}{16} \pi a^4$$

$$77) \int \frac{x}{\sqrt{x^2-7}} dx = \int \frac{\sqrt{7} \sec \theta \sqrt{7} \sec \theta \tan \theta}{\sqrt{7 \sec^2 \theta - 7}} = \int \frac{\sqrt{7} \sec \theta \sqrt{7} \sec \theta \tan \theta}{\sqrt{7(\sec^2 \theta - 1)}} d\theta$$

$$x = \sqrt{7} \sec \theta \quad dx = \sqrt{7} \sec \theta \tan \theta d\theta = \int \frac{\sqrt{7} \sec \theta \sqrt{7} \sec \theta \tan \theta}{\sqrt{7} \tan \theta} d\theta$$

$$\frac{x}{\sqrt{7}} = \sec \theta$$

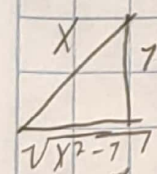
$$\sqrt{7} \int \sec^3 \theta d\theta = \sqrt{7} \tan \theta + C = \sqrt{7} \left(\frac{\sqrt{x^2-7}}{\sqrt{7}} \right)$$



$$\sqrt{x^2-7} + C$$

$$79) \int \frac{\sqrt{1+x^2}}{x} dx = \int \frac{\sqrt{1+\tan^2 \theta}}{\tan \theta} \sec^2 \theta d\theta = \int \frac{\sec \theta \sec^2 \theta}{\tan \theta} d\theta$$

$$x = \tan \theta \quad dx = \sec^2 \theta d\theta = \int \frac{\frac{1}{\cos^3 \theta}}{\frac{\sin \theta}{\cos \theta}} d\theta = \int \frac{\cos \theta}{\sin \theta \cos^2 \theta} d\theta$$



$$= \int \frac{1}{\sin \theta \cos^2 \theta} d\theta = \int \frac{\sec^2 \theta}{\sin \theta} = \frac{1}{\sin \theta} \tan \theta - \int \tan \theta \cot \theta \csc \theta d\theta$$

$$= \frac{\tan \theta}{\sin \theta} - \int \cot \theta \csc \theta \tan \theta d\theta$$

$$u = \frac{1}{\sin \theta} \quad dv = \sec^2 \theta \quad du = -\cot \theta \csc \theta \quad v = \tan \theta = \frac{\tan \theta}{\sin \theta} - \int \frac{1-m^2}{2m} \cdot \frac{1+m^2}{2m} \cdot \frac{2m}{1-m^2} \cdot \frac{2}{1+m^2} dm$$

$$m = \tan\left(\frac{x}{2}\right) \quad \frac{dm}{dx} = \sec^2\left(\frac{x}{2}\right) = \frac{1+\cos x}{\sin x}$$

$$\tan\left(\frac{x}{2}\right) = \frac{1-\cos x}{\sin x}$$

$$= \frac{\tan \theta}{\sin \theta} - \int -\frac{1}{m} dm = \frac{\tan \theta}{\sin \theta} + \ln |\tan\left(\frac{x}{2}\right)|$$

$$= \frac{x}{\sqrt{x^2-1}} + \ln \left| \frac{1-\left(\frac{x}{1}\right)}{\frac{1}{x}} \right| = \frac{x}{\sqrt{x^2-1}} + \ln \left| \frac{1-x}{\frac{1}{x}} \right|$$

$$= \ln \left| \frac{\sqrt{x^2-1}-1}{x} \right| + \sqrt{x^2-1} + C$$

$$a=3 \quad b=5$$

$$21) \int_0^{0.6} \frac{x^2}{\sqrt{9-25x^2}} = \int \frac{\frac{9}{25} \sin^2 \theta}{\sqrt{9-25 \cdot \frac{9}{25} \sin^2 \theta}} = \int \frac{\frac{9}{25} \sin^2 \theta}{\sqrt{9(1-\sin^2 \theta)}} d\theta$$

$$X = \frac{3}{5} \sin \theta \quad dx = \frac{3}{5} \cos \theta$$

$$\frac{x}{\frac{3}{5}} = \sin \theta \quad = 3 \int \frac{\frac{9}{25} \sin^2 \theta}{\sqrt{1-\sin^2 \theta}} d\theta = \frac{27}{25} \int \frac{\sin^2 \theta}{\cos \theta} d\theta$$

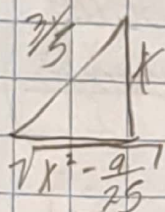
$$= \frac{27}{25} \int \frac{\sin^2 \theta}{\cos \theta} d\theta = \frac{27}{25} \int \frac{\sin^2 \theta \cos \theta}{\cos^2 \theta} d\theta = \frac{27}{25} \int \sin \theta \cos \theta d\theta$$

$$m = \sin \theta$$

$$dm = \cos \theta d\theta$$

$$= \frac{27}{25} \int m \cos \theta \cdot \frac{dm}{\cos \theta} = \frac{27}{25} \int m dm$$

$$\frac{27}{25} \left(\frac{\sin^2 \theta}{2} \right) \Big|_0^{0.6} = \frac{27}{25} \left(\frac{x^2}{\frac{9}{25}} \right) = \frac{27}{25} \left(\frac{25x^2}{9} \right)$$



$$23) \int \frac{dx}{\sqrt{x^2+2x+5}} = \int \frac{dx}{\sqrt{4+(x+1)^2}} = \int \frac{2 \sec^2 \theta}{\sqrt{4+4 \tan^2 \theta}} d\theta = \frac{2}{2} \int \frac{\sec^2 \theta}{\sec \theta} d\theta$$

$$x^2+2x+5$$

$$(x+1)^2 - 1 + 5$$

$$(x+1)^2 + 4$$

$$= \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta|$$

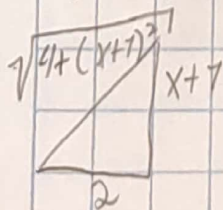
$$= \ln \left| \frac{\sqrt{4+(x+1)^2}}{2} + \frac{x+1}{2} \right| + C$$

$$x+1 = 2 \tan \theta$$

$$dx = 2 \sec^2 \theta d\theta$$

$$= \ln \left| \frac{\sqrt{x^2+2x+5} + x+1}{2} \right|$$

$$\frac{x+1}{2} = \tan \theta$$



$$25) \int x^2 \sqrt{3+2x-x^2} dx = \int x^2 \sqrt{4-(x+1)^2} dx =$$

$$= x^2 - 2x - 3$$

$$= (x-1)^2 - 7 + 3$$

$$= (x-1)^2 - 4 - 3$$

$$= 4 - (x-1)^2$$

$$= \int \frac{4 \sin^2 \theta - 1 \sqrt{4 - 4 \sin^2 \theta}}{4(1 - \sin^2 \theta)} 2 \cos \theta d\theta$$

$$= 16 \int \sin^2 \theta - 1 \cos \theta \cos \theta d\theta$$

$$= 16 \int \cos^4 \theta d\theta = 16 \int \cos^3 \theta \cos \theta d\theta$$

$$\left. \begin{aligned} x+1 &= 2 \sin \theta \\ \frac{x+1}{2} &= \sin \theta \end{aligned} \right\} dx = 2 \cos \theta d\theta$$

$$= 16 \left[\cos^2 \theta \sin \theta - \int \sin \theta - 3 \cos^2 \theta \sin \theta \right]$$

$$u = \cos^3 \theta$$

$$du = -3 \cos^2 \theta \sin \theta$$

$$dv = \cos \theta$$

$$v = \sin \theta$$

$$= 16 \left[\cos^3 \theta \sin \theta + 3 \int \sin^2 \theta \cos^2 \theta d\theta \right]$$

$$= 16 \left[\cos^3 \theta \sin \theta + 3 \left(\frac{1 - \cos 2\theta}{2} \right) \left(\frac{1 + \cos 2\theta}{2} \right) \right]$$

$$= 16 \left[\cos^3 \theta \sin \theta + \frac{3}{4} (1 - \cos 2\theta)(1 + \cos 2\theta) \right]$$

$$= 16 \left[\cos^3 \theta \sin \theta + \frac{3}{4} (1 - \cos^2 2\theta) \right] = 16 \left[\cos^3 \theta \sin \theta + \frac{3}{4} \sin^2 \theta \right]$$

$$= 16 \left[\cos^3 \theta \sin \theta + \frac{3}{4} \frac{1 - \cos 4\theta}{2} \right] = 16 \left[\cos^3 \theta \sin \theta + \frac{3}{8} (1 - \cos 4\theta) \right]$$

$$\left. \begin{aligned} m &= 4\theta \\ dm &= 4d\theta \end{aligned} \right\} = 16 \left[\cos^3 \theta \sin \theta + \frac{3}{8} (1 - \cos m) \frac{dm}{4} \right] = 16 \left[\cos^3 \theta \sin \theta + \frac{3}{32} \theta - \frac{1}{4} \sin 4\theta \right]$$

$$16 \cos^3 \theta \sin \theta + \frac{3}{2} \theta - 4 \sin 4\theta =$$

Answer

$$27) \int \sqrt{5+4x-x^2} dx = \int \sqrt{9-(x-2)^2} dx = \int \sqrt{9-9\sin^2\theta} 3\cos\theta d\theta$$

$$= 3 \int \cos\theta 3\cos\theta d\theta = 9 \int \cos^2\theta d\theta = 9 \int \frac{1+\cos 2\theta}{2} d\theta$$

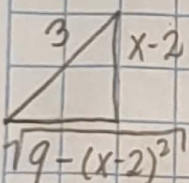
$$= \frac{9}{2} \int 1+\cos 2\theta d\theta = \frac{9}{2} \int 1+\cos m dm = \frac{9}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right)$$

$$x-2 = 3\sin\theta \quad dx = 3\cos\theta d\theta \quad \theta = \sin^{-1}\left(\frac{x-2}{3}\right) + \frac{1}{2} (2\sin\theta\cos\theta)$$

$$\frac{x-2}{3} = \sin\theta \quad m = 2\theta \quad d\theta = \frac{1}{2} dm$$

$$= \sin^{-1}\left(\frac{x-2}{3}\right) + \frac{1}{2} \left(2 \left(\frac{x-2}{3}\right) \left(\frac{\sqrt{9-(x-2)^2}}{3}\right) \right)$$

$$= \frac{9}{2} \sin^{-1}\left(\frac{x-2}{3}\right) + \frac{1}{2} (x-2) \sqrt{5+4x-x^2} + C$$



$$29) \int x \sqrt{1-x^4} dx = \int x \sqrt{1-m^2} \frac{dm}{2x} = \frac{1}{2} \int \sqrt{1-m^2} = \frac{1}{2} \int \sqrt{1-\sin^2\theta} =$$

$$m = x^2 \quad y = \sin\theta \quad dy = \cos\theta d\theta = \frac{1}{2} \int \cos\theta \cos\theta =$$

$$\frac{1}{2} \int \frac{1}{2} (1+\cos 2\theta) d\theta = \frac{1}{4} \int 1+\cos 2\theta$$

$$\frac{1}{4} \left(\theta + \frac{\sin 2\theta}{2} \right) = \theta + \frac{1}{2} \sin\theta \cos\theta = \frac{1}{4} \theta - \frac{\sin\theta \cos\theta}{4}$$

$$\frac{1}{4} \sin^{-1}(m) + \frac{m \sqrt{1-m^2}}{4} = \frac{1}{4} \sin^{-1}(x^2) + \frac{1}{4} x^2 \sqrt{1-x^4} + C$$

37) $\text{ef } X$

$$y = \frac{a}{x^2 + 9}$$

$$y = 0, x = 0, y = 3$$

$$V = \pi x^2 dx$$

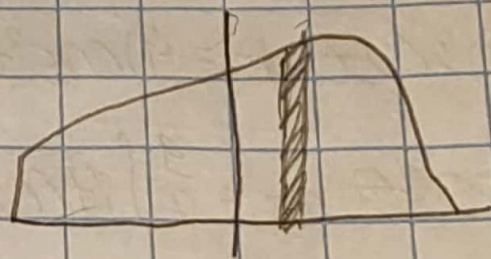
$$V = \pi \int x^2 dx$$

$$V = \pi \int \left(\frac{a}{x^2 + 9} \right)^2 dx$$

$$V = \int_0^3 \frac{81}{(x^2 + 9)^2} dx$$

$$m = x^2$$

$$dm = 2x dx$$



~~Do~~ Doda